SHADRACK ITULAH

Software Engineer | MSc Computer Science
Hatfield, United Kingdom
+447833825441 | shedrackitulah@gmail.com | LinkedIn | Portfolio GitHub

Education

University of Hertfordshire — Hatfield, UK
Master of Science in Computer Science — *2025 - 2027*

Obafemi Awolowo University — Nigeria
Bachelor of Science in Agricultural and Environmental Engineering — *2017 - 2022*

Certifications

AWS Certified Cloud Practitioner — 2026
Amazon Web Services (AWS)

Machine Learning with Python — 2026
IBM

Deep Learning with Keras and TensorFlow — 2026
IBM

Advanced RAG with Vector Databases and Retrievers — 2026
IBM

Agentic AI with LangGraph, CrewAI, AutoGen and BeeAI — 2026
IBM

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL
Computer Science: Data Structures and Algorithms, Object-Oriented Design, Complexity Analysis, Operating Systems, System Design, Distributed Systems
Backend: FastAPI, Node.js, Express.js, REST APIs, WebSockets, Server-Sent Events, Celery
Databases: PostgreSQL, MongoDB, Redis, Qdrant
Cloud/DevOps: AWS (EC2, S3, Lambda), Docker, GitHub Actions, CI/CD
Testing: Pytest, Jest, Vitest, Testcontainers, Unit, Integration and End-to-End Testing
AI/Retrieval: RAG, Hybrid Search, Vector Search, Embeddings, Cross-Encoder Reranking, Ollama

Experience

Full-Stack Engineer — 2024 – 2025
Techinnover

- Architected and developed a production cinema booking platform using **Next.js**, **FastAPI**, and **PostgreSQL**, supporting 10,000+ monthly users with real-time seat availability updates through **WebSockets**.
- Designed and implemented **RESTful APIs** for authentication, seat allocation, payments, and booking workflows, integrating Stripe and supporting 1,000+ daily transactions.
- Optimized PostgreSQL queries using **SQLAlchemy** and introduced **Redis caching**, reducing API response times by 25% and improving performance across high-traffic application workflows.
- Designed containerized services with **Docker** and deployed applications on **AWS EC2**, implementing automated build, test, and deployment workflows with **GitHub Actions CI/CD**.
- Owned features from technical design through implementation, testing, code review, and production deployment; delivered 15+ features, participated in 130+ code reviews, and mentored 3 junior developers.

Full-Stack Engineer — 2023 – 2024
Monomango (Web Development Agency)

- Built and deployed 8+ production applications using **React**, **TypeScript**, **Node.js**, and **PostgreSQL** across e-commerce, healthcare, and education products.
- **Integrated AI-powered search functionality** using vector embeddings and semantic search, improving search accuracy by 45% and user engagement by 30% through intelligent query understanding.
- Collaborated on **database schema design** and optimized complex SQL queries for high-traffic endpoints serving 100K+ monthly users, improving application responsiveness and scalability.

- Designed secure **RESTful APIs** with Express.js, implementing authentication, authorization, input validation, and reusable middleware for application services.
- Implemented Redis caching, automated testing with **Jest** and **Supertest**, and CI/CD workflows that reduced deployment time from 2 hours to 15 minutes.

Frontend Engineer 2022 – 2023

Bx Studio (Web Development Agency)

- Led frontend development across 12+ client projects using **React**, JavaScript, and Webflow, building responsive interfaces and reusable components for production applications.
- Developed custom backend integrations using **Xano** as an API and data layer, implementing application logic and data models supporting 50K+ records across multiple products.
- Integrated third-party APIs and services including **Stripe**, Google Maps, and SendGrid, connecting frontend workflows with payments, geolocation, and transactional communication services.

Featured Projects

DocRAG – Local-First RAG Knowledge | *FastAPI, PostgreSQL, Qdrant, Redis, Celery, Docker* 2026

- Architected a **local-first RAG platform** with project-isolated knowledge bases and asynchronous document ingestion pipelines, processing PDF, DOCX, PPTX, HTML, Markdown, and text files using **Celery**, PostgreSQL, S3-compatible object storage, and Qdrant.
- Implemented **hybrid information retrieval** combining dense semantic embeddings and BM25-style sparse search, merging ranked results with **Reciprocal Rank Fusion (RRF)** and applying cross-encoder reranking to improve retrieval relevance.
- Developed citation-aware RAG chat with **streamed responses**, metadata-filtered retrieval, page-level source attribution, and strict user/project retrieval boundaries to prevent cross-project data access.
- Built reliable ingestion and retrieval infrastructure with **Redis**, Server-Sent Events, database migrations, document cleanup across relational, vector, and object storage layers, and automated testing using **Pytest**, Testcontainers, strict mypy, and GitHub Actions CI.
- Integrated **BEIR-based retrieval evaluation** to benchmark dense, sparse, hybrid, and reranked search pipelines using standard information-retrieval metrics, enabling quantitative comparison of retrieval strategies and regression testing of search quality.

AI-Powered Data Analysis Platform | *Next.js, FastAPI, PostgreSQL, LangGraph, Docker* 2026

- Designed and built a full-stack data analysis platform using **Next.js**, **FastAPI**, and **PostgreSQL**, enabling users to upload structured datasets and execute natural-language analysis workflows through typed backend APIs.
- Implemented multi-stage analysis workflows with **LangGraph**, coordinating query interpretation, dataset inspection, analysis planning, code generation, execution, and response validation across reusable stages.
- Built a secure **Python execution sandbox** for generated analysis code, supporting Pandas-based transformations and Plotly visualizations while isolating execution from the core application.
- Containerized the application with **Docker** and implemented authenticated API access using JWT access and refresh token flows, separating frontend, backend, database, and execution responsibilities for reliable deployment.